

MICROBIOLOGY – SHORT EXAM NOTES

Hypersensitivity (Ananthanarayan & Paniker, Ch. 16)

1. INTRODUCTION AND CLASSIFICATION

Hypersensitivity refers to the **undesirable, injurious consequences in a sensitised host** following contact with a specific antigen – an **exaggerated or inappropriate immune response** leading to tissue damage, disease or even death. In immunity the focus is on what happens to the antigen; in hypersensitivity the antigens are often **innocuous** (pollen, serum proteins) and the focus is on what happens to the **host**. **Allergy** refers to all harmful immune processes (hypersensitivity and autoimmunity) and is commonly used as a synonym for hypersensitivity.

Hypersensitivity requires **prior contact** with the antigen (**allergen**): the first contact is the **sensitising (priming) dose**; later contact is the **shocking dose**, which produces the reaction.

Immediate vs Delayed Hypersensitivity

Immediate hypersensitivity (B cell / antibody mediated)	Delayed hypersensitivity (T cell mediated)
Appears and recedes rapidly (minutes)	Appears slowly (24–72 h), lasts longer
Induced by antigens or haptens by any route	Antigen intradermally, with Freund's adjuvant or by skin contact
Circulating antibodies present and responsible	Antibodies may be absent; cell-mediated
Passive transfer with serum possible	Transfer only with T cells / transfer factor , not serum
Desensitisation easy but short-lived	Desensitisation difficult but long-lasting
Anaphylaxis, atopy, antibody-mediated cell damage, Arthus reaction, serum sickness	Infection (tuberculin) type, contact dermatitis

Coombs and Gell Classification (1963)

Type	Mechanism	Mediators	Time	Examples
I – Anaphylactic (IgE, reagin)	Antigen binds IgE fixed on mast cells/basophils → release of vasoactive amines	IgE ; histamine, leukotrienes, PGs	Minutes	Anaphylaxis, atopy (hay fever, asthma), insect sting, drug allergy
II – Cytotoxic / cytolytic	IgG/IgM against cell-surface or tissue antigens → complement lysis or cytotoxic cells	IgG, IgM, complement	Hours–days	Transfusion reactions, haemolytic disease of newborn , drug-induced haemolytic anaemia, agranulocytosis, thrombocytopenia
III – Immune complex	Antigen–antibody complexes deposit in/around vessels → complement, neutrophils	IgG, IgM, complement, leucocytes	Hours–days	Arthus reaction, serum sickness , glomerulonephritis, farmer's lung
IV – Delayed (cell-mediated)	Sensitised CD4 and CD8 T cells release lymphokines, macrophages accumulate	T cells, lymphokines, macrophages	24–72 h	Tuberculin test, contact dermatitis , granulomas, graft rejection

Type	Mechanism	Mediators	Time	Examples
V – Stimulatory	Antibody binds cell-surface receptor and stimulates it (modified type II)	Antibody	–	LATS in Graves' disease

The mechanisms are **not mutually exclusive** – a reaction may involve more than one type, and non-immune systems (inflammation, complement, coagulation, kinins – the **humoral amplification system**) modify the clinical picture.

2. TYPE I HYPERSENSITIVITY (IgE DEPENDENT)

Occurs in two forms: the **acute, potentially fatal, systemic** form – **anaphylaxis**; and the **chronic or recurrent, non-fatal, localised** form – **atopy**.

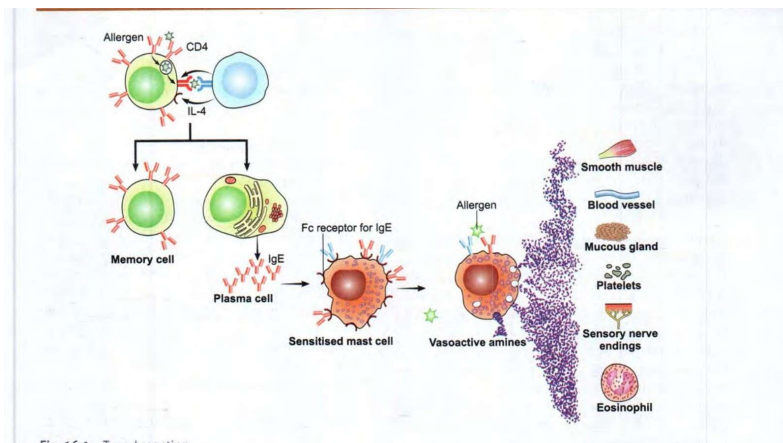


Fig. 16.1 Type I reaction (from the book)

Anaphylaxis

The **classical immediate hypersensitivity reaction**.

- Induced by antigens or haptens; an interval of **at least 2–3 weeks** is needed between sensitising and shocking dose; sensitisation lasts long.
- Shocking dose is most effective **intravenously**, less intraperitoneally or subcutaneously, least intradermally; it must be identical or closely related to the sensitising antigen.
- Clinical effects are the same for any antigen but vary between species; due to **smooth muscle contraction and increased vascular permeability**. The organs chiefly involved are '**shock organs**' / **target tissues**.
- In humans**: itching of scalp and tongue, flushing, **difficulty in breathing (bronchospasm)**, nausea, vomiting, abdominal pain, diarrhoea, **acute hypotension**, loss of consciousness and death. Now seen mostly after **injection of antibiotics or drugs** and **insect stings**.
- Treatment**: prompt **adrenaline 0.5 ml of 1:1000 solution subcutaneously or intramuscularly** can be life-saving; antihistamines.

Types of anaphylaxis

- Cutaneous anaphylaxis** – small intradermal shocking dose gives a local **wheal-and-flare**; used as a **skin test** for allergy (keep adrenaline ready).
- Passive cutaneous anaphylaxis (PCA)** – Ovary (1952); antibody injected intradermally, then antigen + **Evans blue IV** → blueing at the site; very sensitive test for antibodies (detects IgG, not IgE).

- **Anaphylaxis in vitro – Schultz–Dale phenomenon** – intestinal or uterine muscle strips from a sensitised guinea pig contract on adding the specific antigen.

Mechanism

The basis is **cytotropic IgE** antibody (IgG1 in guinea pig and mouse); circulating free IgE is not relevant. IgE binds by its **Fc end to FcεR receptors on mast cells and basophils**. On the shocking dose, antigen **cross-links adjacent cell-bound IgE**, increasing calcium influx and causing **degranulation** with release of mediators.

Mediators of anaphylaxis

Mediator	Source / nature	Action
Histamine (most important in humans)	Preformed; mast cells, basophils, platelets (from histidine)	Itching; flare (vasodilatation) and wheal (increased permeability); smooth muscle contraction
Serotonin (5-HT)	From tryptophan; intestine, brain, platelets	Smooth muscle contraction, increased permeability, vasoconstriction
ECF-A / NCF	Mast cell granules	Chemotaxis of eosinophils (eosinophilia) and neutrophils
Leukotrienes (SRS-A: LTB4, C4, D4, E4)	Newly formed – lipoxygenase pathway	Slow, sustained smooth muscle (bronchial) contraction
Prostaglandins, thromboxane A2	Cyclo-oxygenase pathway	Bronchoconstriction, platelet effects, pain
Platelet activating factor (PAF)	Basophils	Platelet aggregation and amine release
Others	Complement, plasma kininogens	Anaphylatoxins (C3a, C5a), bradykinin

Anaphylactoid reaction – a clinical reaction resembling anaphylaxis caused by IV peptone, trypsin etc.; it has **no immunological basis** (non-specific complement activation and anaphylatoxin release).

Atopy

Term introduced by **Coca (1923)** for **naturally occurring familial hypersensitivities** of humans, e.g. **hay fever, asthma**, allergic rhinitis, food allergy, urticaria and atopic dermatitis.

- Allergens (**atopens**) are **inhalants** (pollen, house dust) or **ingestants** (eggs, milk); poor antigens when injected but induce **IgE (formerly 'reagin')**; sensitisation occurs spontaneously by natural contact.
- **Genetically determined** – what is inherited is the tendency to produce **large amounts of IgE** (about **10%** of people); runs in families; hard to induce experimentally.
- **IgE is heat-sensitive** (inactivated at 56°C in 2–4 h), **does not cross the placenta**, and is **homocytotropic** (fixes only to cells of the same species).
- **Prausnitz–Küstner (PK) reaction (1921)** – serum of an atopic person injected intradermally into a normal person, followed 24 h later by the antigen at the same site, gives a wheal-and-flare; the original test for atopic antibody; no longer used (risk of infection). Now IgE is measured by **ELISA**, **RAST**, etc.
- Often associated with **IgA deficiency** (mucosal IgA normally prevents allergens reaching IgE-forming cells).

- Clinical features depend on the **portal of entry** – conjunctivitis, rhinitis, GI symptoms, dermatitis, or remote effects like urticaria. Treated by **specific desensitisation (hyposensitisation)**, antihistamines.

Diagnostic tests for type I: total IgE levels, direct skin (prick) test, **radioallergosorbent test (RAST)**, leucocyte histamine release test, provocation tests.

3. TYPE II – CYTOLYTIC AND CYTOTOXIC REACTIONS

IgG (or rarely IgM) antibodies combine with antigens **on the cell surface**, leading to cell damage by **complement-mediated lysis, opsonisation and phagocytosis**, or **antibody-dependent cytotoxicity by K (killer) cells**. The antigen may be a native cell antigen (e.g. RBC antigens) or a **drug/hapten adsorbed** on cells. Type II is intermediate between hypersensitivity and autoimmunity.

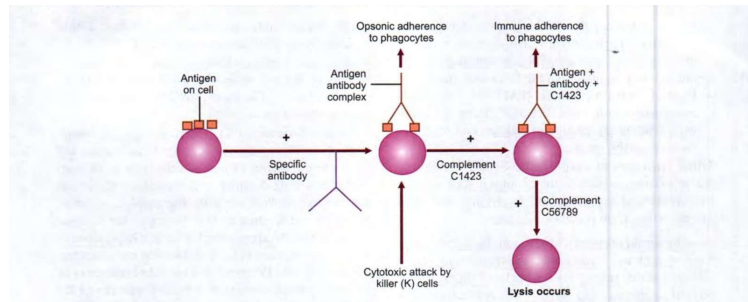


Fig. 16.2 Type II reaction (from the book)

Examples: incompatible blood transfusion reactions, haemolytic disease of the newborn (Rh incompatibility), autoimmune haemolytic anaemia, drug-induced haemolytic anaemia, agranulocytosis and thrombocytopenic purpura, Goodpasture's syndrome.

4. TYPE III – IMMUNE COMPLEX DISEASES

Damage is caused by **antigen–antibody complexes** that precipitate in and around **small blood vessels** or on membranes. Complexes activate **complement (classical pathway)** – C3a and C5a attract **neutrophils**, which release **lytic enzymes**; histamine increases permeability, and platelet–leucocyte thrombi cause necrosis. Complement levels in plasma fall. The disease is usually self-limiting once antigen is eliminated.

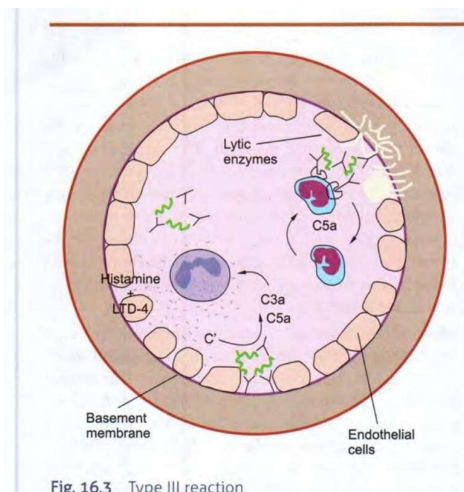


Fig. 16.3 Type III reaction

Fig. 16.3 Type III reaction (from the book)

Arthus Reaction (Local Type III)

Arthus (1903) – rabbits repeatedly injected **subcutaneously with horse serum**: early injections had no effect, but later injections produced an intense **local reaction with oedema, induration and haemorrhagic necrosis**. Due to **antigen–antibody precipitates** at the site with complement activation, neutrophil

infiltration and thrombosis. Can be passively transferred with serum containing precipitating **IgG/IgM** in high titre. **Clinical counterpart: farmer's lung** (thermophilic actinomycetes in mouldy hay) and other **hypersensitivity pneumonitis**.

Serum Sickness (Systemic Type III)

Described by **von Pirquet and Schick (1905)**; appears **7–12 days after a single injection** of a large dose of foreign serum (e.g. **diphtheria antitoxin**), now more often after **penicillin** and other drugs. Features: **fever, lymphadenopathy, splenomegaly, arthritis, glomerulonephritis, endocarditis, vasculitis, urticarial rashes**, abdominal pain, nausea, vomiting. Antibody formed against the persisting foreign serum makes complexes that deposit on vessel walls. **Unique feature: a single injection** acts as both sensitising and shocking dose. Other type III conditions: post-streptococcal glomerulonephritis, hepatitis B, malaria, filaria and certain autoimmune diseases.

5. TYPE IV – DELAYED HYPERSENSITIVITY

An aspect of **cell-mediated immunity**, typically provoked by **intracellular infections** or **haptens (chemicals) on skin**. It evolves slowly (**24–72 h**) with a mixed infiltrate of **lymphocytes and macrophages**. It is caused not by antibody but by **sensitised T cells** (TDTH, TH1, TH2, Tc) which on meeting antigen release **cytokines**. It **cannot be transferred by serum** but can be by **lymphocytes or transfer factor**.

1. Tuberculin (Infection) Type

The archetype. Intradermal **tuberculin (PPD)** in a person sensitised by prior infection or BCG produces an **indurated inflammatory reaction in 48–72 hours**; no reaction in unsensitised persons. Indicates the state of **CMI** to tubercle bacilli. Develops in many subacute or chronic infections with intracellular bacteria, fungi, viruses and parasites, and in **allograft rejection** and autoimmune diseases.

2. Cutaneous Basophil Hypersensitivity (Jones–Mote Reaction)

A tuberculin-like local reaction to intradermal protein antigens with prominent **basophil infiltration**; **can be transferred by serum**, so it is not a true delayed reaction; significance unknown.

3. Contact Dermatitis Type

Follows skin contact with chemicals – metals (**nickel, chromium**), dyes, picryl chloride, dinitrochlorobenzene, drugs (penicillin, antibiotic ointments), plant allergens (**parthenin** from parthenium, poison ivy) and toiletries. These act as **haptens** combining with skin proteins. **Mechanism: Langerhans cells** capture the hapten–protein, migrate to lymph nodes and present it with MHC to T cells; sensitised T cells return to the skin – **TH1** release **IFN- γ and IL-2** (activate macrophages), **TH2** release IL-4, IL-5, GM-CSF (eosinophils), **Tc** kill target cells. **Lesions:** macules, papules, vesicles that break into raw weeping areas (**acute eczematous dermatitis**). **Diagnosis: patch test** – allergen under an adhesive dressing; itching in 1–5 h and erythema to vesicles at 24–48 h.

6. TYPE V AND SHWARTZMAN REACTION

Type V (stimulatory hypersensitivity): antibody binds to a **cell-surface receptor** and **stimulates** the cell instead of destroying it – e.g. **long-acting thyroid stimulator (LATS)** against thyroid cells in **Graves' disease** causing excess thyroid hormone; also cited: Stevens–Johnson syndrome, sulphonamide-induced morbilliform rash.

Shwartzman reaction – **not an immune reaction** but a disturbance of intravascular coagulation. **Shwartzman (1928):** intradermal injection of *S. Typhi* culture filtrate (**endotoxin – preparatory dose**) in a rabbit, followed **24 h later** by an **intravenous provocative dose**, produces a **haemorrhagic necrotic lesion**

at the skin site. The two doses need not be related, so it is **non-specific**. If both doses are IV, the animal dies in 12–24 h – **generalised (Sanarelli–Shwartzman) reaction**. **Mechanism**: leucocytes at the prepared site damage capillaries; the second dose causes intravascular clotting (alternative complement activation, thromboxane, TNF, IL-1) → **DIC**. Clinical parallels: **purpuric rashes of meningococcal septicaemia** and **Waterhouse–Friderichsen syndrome** (haemorrhagic adrenal necrosis).

★ **EXAM FOCUS**

- Essay: **Classify hypersensitivity reactions with examples. Describe type I hypersensitivity.**
- Short answers: **Anaphylaxis, Atopy, Prausnitz–Küstner reaction, Arthus reaction, Serum sickness, Tuberculin type DH, Contact dermatitis, Shwartzman reaction, Delayed hypersensitivity, Mediators of anaphylaxis.**
- One-liners: type I – IgE, mast cells, minutes; type II – IgG/IgM on cell surface (transfusion reaction, HDN); type III – immune complexes (Arthus, serum sickness 7–12 days); type IV – T cells, 48–72 h (Mantoux, patch test); type V – LATS; adrenaline 1:1000, 0.5 ml IM.